XFOIL Guide – AE4130 Assignment Part 1

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How to run every XFOIL part of the AE4130 Part 1 assignment: the exact command sequences, what each output file contains, and what to plot. Examples use NACA 1611; replace it with your group's airfoil.

Setup

XFOIL is a text-command program: you type commands at a prompt, it prints results and draws plots in a separate window. Your report plots come from the files it writes, not from that window.

1. Download the Windows version (6.99) from the Brightspace link or the XFOIL home page and unzip it.
2. Make a working folder **with no spaces in the path**, e.g. `...\AircraftAerodynamics\XFOIL_runs`, and copy `xfoil.exe` into it. XFOIL writes every output file into the folder it was started from.
3. Double-click `xfoil.exe`. A console window opens with the prompt `XFOIL c>`.

How the menus work

- XFOIL has a top level and sub-menus. The prompt tells you where you are: `XFOIL c>` (top), `.OPERi c>` (analysis, **i = inviscid**), `.OPERv c>` (analysis, **v = viscous**), `.PPAR`, `.VPAR`, `.GDES`, `.MDES`.
- Type a command and press Enter. Commands are not case-sensitive. Many take values on the same line (`ALFA 5`); if you leave them out, XFOIL asks.
- **An empty Enter takes you up one level.** Press it repeatedly to get back to the top.
- `?` lists the commands of the current menu.
- File names: short, no spaces, ideally with an extension (`cp_a10.txt`).
- Writing to a file name that already exists: XFOIL asks before overwriting, or for polars, may refuse (see Output files).

If the plot window gets in the way, it is safe to ignore. `PLOP` → `G` at the top level toggles graphics off.

Tip: the c> prompt doubles as your check. If you expect a viscous result and see `.OPERi`, viscous mode is off.

Command cheat sheet

Menu	Command	What it does
Top	`NACA 1611`	Creates a NACA 4-digit airfoil, paneled with the current settings (160 nodes by default)
Top	`PPAR`	Paneling menu. `N 50` sets the number of panel nodes; empty Enter regenerates and exits
Top	`SAVE name.dat`	Writes the current airfoil coordinates
Top	`LOAD name.dat`	Reads an airfoil from a coordinate file
Top	`PANE`	Re-panels the buffer airfoil into the current one (needed after MDES)
Top	`GDES`	Geometry menu (thickness scaling)
Top	`MDES`	Full-inverse design menu
Top	`OPER`	Analysis menu
OPER	`VISC 1.5e6`	**Toggles** viscous mode on (with this Re). Typing `VISC` again switches it off
OPER	`RE 3e5`	Changes Re while staying viscous
OPER	`MACH 0`	Mach number (keep 0)
OPER	`ITER 100`	Raises the viscous iteration limit (default is low)
OPER	`INIT`	Resets the boundary layer (after a failed point)
OPER	`VPAR`	BL settings menu: `N 12` = Ncrit; `XTR 0.3 1` = forced transition top, bottom
OPER	`ALFA 5`	Solves at one angle of attack
OPER	`ASEQ -6 18 0.5`	Solves a sweep of angles
OPER	`CL 0.4`	Solves at a fixed lift coefficient
OPER	`PACC`	Starts / stops saving every converged point to a polar file
OPER	`CPWR name.txt`	Writes Cp along the surface for the current point
OPER	`DUMP name.txt`	Writes boundary-layer data (incl. Cf) for the current point

Menu	Command	What it does
OPER	`CPX` / `VPLO`	Screen plots of Cp / BL quantities (for checking only)

Two things catch everyone out:

- `VISC` is a toggle. If you are already viscous and type `VISC 3e5` to change Re, you switch viscous **off**. Use `RE` instead.
- `CPWR` and `DUMP` write the point you solved **last**. Solve the angle you want first, then write.

Output files

Open every file type in a text editor once before writing your script, and read its header: column layouts differ slightly between XFOIL versions.

File	Made by	Columns (6.99)	Watch out for
Polar	`PACC` + `ASEQ` / `ALFA` / `CL`	alpha, CL, CD, CDp, CM, Top_Xtr, Bot_Xtr (+ Top_Itr, Bot_Itr)	Points are stored in the order you ran them, not sorted by α. Unconverged points are simply missing. Inviscid: CD = 0
Cp	`CPWR`	x, (y,) Cp	Runs from the trailing edge over the upper surface to the leading edge, then back along the lower surface. Split at the minimum x
BL dump	`DUMP`	s, x, y, Ue/V∞, Dstar, Theta, Cf, H, …	Same surface order as Cp, followed by **wake** points (x > 1). Drop the wake
Coordinates	`SAVE`	x, y	Use for the airfoil-shape figure

How `PACC` behaves. It asks two things: a polar save file name (type one) and a polar dump file name (just press Enter). From then on every converged point is appended. Type `PACC` again to stop. If the file exists and was made with the same airfoil and conditions, XFOIL appends to it; if anything differs, it refuses to write so the old polar is not overwritten. **Use a new file name for every new configuration.**

Suggested naming: `pol_1611_inv.txt`, `pol_1611_N50_inv.txt`, `pol_1611_Re1.5e6.txt`, `cp_1611_a10_inv.txt`, `dump_1611_a4.txt`. Files then explain themselves in your script and in your folder.

Theme 1 — inviscid questions

XFOIL starts **inviscid** (`.OPERi`), so Theme 1 needs no extra settings. 3.1.1 is theory only. In the sequences below, `↵` means press Enter on an empty line. Every sequence starts at the top level (`XFOIL c>`): if you are still in OPER, press Enter first.

Airfoils you will need

Label	Rule	Example (NACA 1611)
Original ABCD	Group table	1611
Thickness variant ABEF (3.1.3)	EF = 2·CD if CD ⩽ 14; EF = CD/2 if CD > 14 (the brief's "ED" is a typo for EF)	1622
Symmetric 00CD (3.1.6)	Same thickness, no camber	0011

3.1.2 — Cp at 0°, 5°, 10°

```
NACA 1611
OPER
ALFA 0
CPWR cp_1611_a0_inv.txt
ALFA 5
CPWR cp_1611_a5_inv.txt
ALFA 10
CPWR cp_1611_a10_inv.txt
```

Plot all three on one figure, Cp axis inverted. Discuss the suction peak, the stagnation point moving onto the lower surface, and the area between the curves (which is Cl).

3.1.3 — thickness variant at 10°

```
↵  (back to top level)
NACA 1622
OPER
ALFA 10
CPWR cp_1622_a10_inv.txt
```

Plot it against `cp_1611_a10_inv.txt`.

3.1.4 — Cl–α and Cm–α, −6° to 18°

For each of the two airfoils:

```
NACA 1611
OPER
PACC
pol_1611_inv.txt
↵
ASEQ -6 18 0.5
PACC
```

Repeat with `NACA 1622` and `pol_1622_inv.txt`. Expect straight Cl lines with no stall; inviscid flow cannot separate. CM in the file is about c/4.

Experimental data: Abbott & von Doenhoff, *Theory of Wing Sections*, Appendix IV, covers only some 4-digit sections (the 00xx, 14xx, 24xx and 44xx families). A 16xx section is probably not in it: say so in the report and compare with the nearest section that is (same camber or same thickness). Airfoiltools.com polars are XFOIL results, not experiments.

Stability: look at dCm/dα and the sign of Cm at zero lift. c/4 is close to the aerodynamic centre, so Cm,c/4 is nearly flat.

3.1.5 — panel study (10, 50, 250)

XFOIL's `N` is the number of panel nodes, and its default of 160 is the brief's "160 panels". So use N = 10, 50 and 250.

```
NACA 1611
PPAR
N 10
↵  (repeat until you see XFOIL c>)
OPER
ALFA 10
CPWR cp_1611_N10_a10.txt
PACC
pol_1611_N10_inv.txt
↵
ASEQ -6 18 0.5
PACC
```

Repeat with `N 50` and `N 250`. Plot Cp for 10, 50, 160 and 250 with markers so the 10-node points are visible. Compare Cl and Cm at a few angles in a small table to justify your minimum.

3.1.6 — symmetric foil vs thin-airfoil theory

Make sure N is back at 160 (`PPAR` → `N 160`), then:

```
NACA 0011
OPER
PACC
pol_0011_inv.txt
↵
ASEQ -6 18 0.5
PACC
```

Thin-airfoil theory for a symmetric section gives Cl = 2πα (α in rad) and Cm,c/4 = 0. Compute that line yourself and overlay it. XFOIL's slope comes out slightly above 2π, because thickness adds lift that thin-airfoil theory ignores.

Theme 2 — viscous polars, transition and Cf

The Prandtl equations, the e^N method and the stability diagram are theory and hand drawings. Everything below uses the original airfoil at 160 nodes, **Re = 1.5 × 10⁶, Mach 0, Ncrit 9** (XFOIL's default; check with `VPAR` if you changed it).

Lift and drag polar, −6° to 18°

Sweep **outward from 0°** in two parts. The viscous solver uses each converged point as the starting guess for the next, so starting near zero lift converges far better than starting at −6°.

```
NACA 1611
OPER
VISC 1.5e6
ITER 100
PACC
pol_1611_Re1.5e6.txt
↵
ASEQ 0 18 0.5
INIT
ASEQ 0 -6 -0.5
PACC
```

- The prompt should now read `.OPERv` .
- `INIT` resets the boundary layer before the second sweep (in older XFOIL builds it sits inside `VPAR`).

- The file holds both sweeps, so α = 0 appears twice and the order is mixed: sort by α in your script.
- Points that did not converge are missing; near stall that is normal. See Troubleshooting before accepting large gaps.

Plot Cl–α, Cl–Cd and Cm–α against your inviscid polar from 3.1.4.

Transition location vs α (0° to 10°)

No new run: plot the `Top_Xtr` column of `pol_1611_Re1.5e6.txt` against α for 0–10°. XFOIL also prints the transition location after every point ("Side 1 free transition at x/c = …"; side 1 is the upper surface).

Cf along the suction side at α = 0° and 4°

Still in `.OPERv` with the same settings:

```
ALFA 0
DUMP dump_1611_a0.txt
ALFA 4
DUMP dump_1611_a4.txt
```

Keep the upper-surface rows (from the first row to the leading edge, where x is smallest) and plot Cf against x. The jump in Cf marks transition and should match `Top_Xtr`: a quick check that you picked the right surface. `VPLO` → `CF` shows the same on screen.

Cl/Cd at fixed Cl = 0.4

```
CL 0.4
```

XFOIL prints α, CL, CD, CM and the transition locations. Compute Cl/Cd, and note Top_Xtr: it is the baseline for the redesign. To keep a record in a file, wrap it in `PACC` with a new file name (e.g. `pol_1611_Cl0.4.txt`).

Inverse design (MDES) and thickness matching

In MDES you change the upper-surface **surface-speed distribution Qspec** at your design condition, and XFOIL computes the geometry that produces it. The aim: keep the flow accelerating further aft, which moves transition back, extends laminar flow and lowers Cd at Cl = 0.4.

What a good target looks like

- **Original:** speed peaks near the leading edge, then decelerates over most of the chord. The adverse gradient triggers early transition.
- **Target:** a lower leading-edge peak, then speed rising slowly to a maximum around 40–60 % chord, then a smooth recovery to the trailing edge.
- **Don't overdo the recovery.** A short, steep deceleration near the trailing edge can cause turbulent separation, which costs more drag than the laminar flow saves. Make small changes and re-analyse often.

Design

```
NACA 1611
MDES
CQ 0.4
MODI
SMOO
EXEC
↵
PANE
SAVE 1611_opt1.dat
```

- `CQ 0.4` shows Qspec at Cl = 0.4 (inviscid).
- `MODI` works in the plot window: click new points along the upper-surface Qspec curve, from one end of the region you want to change to the other. XFOIL fits a smooth curve through them and changes only that region. The console tells you how to finish (right-click or a key press, depending on the build).
- `SMOO` smooths the modified segment; use it if the curve looks wavy.
- `EXEC` computes the new shape, but only as the *buffer* airfoil. **`PANE` at the top level makes it the current airfoil**; without it, OPER still analyses the old one.
- Check the new shape on screen: no wiggles, no crossed trailing edge. If it looks wrong, go back into `MDES` and make a smaller change.

Match the thickness

The brief requires the same t/c as the original (e.g. 11 % for a xx11). XFOIL prints max thickness and camber when it panels an airfoil, so read the new value after `PANE`, then scale:

```
GDES
TFAC 0.95 1
EXEC
↵
PANE
SAVE 1611_opt1_t11.dat
```

`TFAC` takes a thickness factor and a camber factor. Set the thickness factor to t_target / t_current (0.95 is only an example) and the camber factor to 1 so camber is unchanged. Check the printed thickness again afterwards.

Evaluate and iterate

```
OPER
VISC 1.5e6
ITER 100
CL 0.4
CPWR cp_opt1_Cl0.4.txt
```

Check the prompt says `.OPERv` (VISC is a toggle). Compare CD, Cl/Cd and Top_Xtr with the original at Cl = 0.4. If it didn't improve, `LOAD` an earlier version and try a gentler change. For the report: both shapes from the `SAVE` files (equal axis scaling), Cp of both at Cl = 0.4 on one figure, and a table of α, Cd, Cl/Cd and Top_Xtr. Keep every version (`_opt1`, `_opt2`, …).

Laminar separation bubble and forced transition

The questions on why lift drops and drag rises, and on critical roughness and zig-zag tape, are theory and hand drawings. The XFOIL work: find a bubble, show how its length changes with Re, and remove it with a trip. All runs use **Ncrit = 12**.

Find a Reynolds number with a clear bubble

```
NACA 1611
OPER
VISC 3e5
VPAR
N 12
↵
ITER 100
ALFA 4
CPWR cp_1611_Re3e5_a4.txt
DUMP dump_1611_Re3e5_a4.txt
```

What a bubble looks like:

- **In Cp (upper surface):** a **plateau**, where Cp stays almost constant over a stretch of chord because the separated laminar layer cannot recover pressure, then a **sharp recovery** where the flow transitions and reattaches. `CPX` on screen shows the viscous Cp next to the inviscid one, so the plateau stands out where they part.
- **In the dump: Cf ⩽ 0** between separation and reattachment (reversed flow). Top_Xtr sits inside the plateau, near its end.

If the bubble is weak, try another α (2–6°) or Re in the brief's range (2–6 × 10^5). Annotate separation, transition and reattachment on the Cp plot you use.

Bubble length vs Re

Keep α fixed and change Re with `RE` (not `VISC`, which would switch viscous off):

```
RE 2e5
INIT
ALFA 4
CPWR cp_1611_Re2e5_a4.txt
DUMP dump_1611_Re2e5_a4.txt
```

Repeat for 4 × 10^5 and 6 × 10^5. Read the start and end of Cf < 0 in each dump and overlay the Cp curves.

Remove the bubble with a trip

1. Choose a fixed Cl (e.g. the Cl of your bubble case). Run it untripped with `CL <value>`, note CD and the **laminar separation point** (where Cf first reaches 0).
2. Set the upper-surface trip just upstream of it, keeping the lower surface free:

```
VPAR
XTR 0.30 1
↵
INIT
CL 0.6
CPWR cp_1611_Re3e5_trip0.30.txt
```

(0.30 and 0.6 are examples; use your own separation point and Cl.)

3. The plateau should be gone and CD lower. Try two or three trip positions: too far forward adds turbulent skin friction over a longer stretch, so CD rises again; too far aft lets the bubble re-form. Report the best one next to the untripped case.
4. Reset with `XTR 1 1` in `VPAR` afterwards.

Troubleshooting

Symptom	Likely cause	Fix
“VISCAL: Convergence failed”	Viscous solver did not converge in the iteration limit	`ITER 100` – `200`; `INIT`; approach the angle in smaller steps from a converged point
Many points missing near stall	Normal for XFOIL post-stall	Smaller step (0.25°) over that range; accept a few gaps and say so
CD = 0 in a Theme 2 polar	Viscous mode off (`.OPERi`)	`VISC 1.5e6`; check the prompt
Viscous switched off unexpectedly	Typed `VISC` while already viscous	Use `RE` to change Re
Polar file empty or not updated	File existed with different settings, or `PACC` not on	New file name; `PACC` before the sweep
Results still from the old airfoil after MDES/GDES	Buffer not made current	`EXEC` in GDES, then `PANE` at the top level
`NACA` gives 10 nodes when you wanted 160	PPAR N still set from the panel study	`PPAR` → `N 160`
Can't find output files	XFOIL writes to the folder it started in	Start `xfoil.exe` from your working folder

Before trusting a viscous point, check that it converged (the console says so) and that transition in the polar matches the jump in Cf.

Sources

- AE4130 Assignment Part 1 brief (2026/27).
- XFOIL 6.9 User Primer, M. Drela & H. Youngren (PDF): menus, `VISC` toggle, `PACC` file behaviour, MDES commands and the `PANE` step after `EXEC`.
- XFOIL home page (MIT): download and `xfoil_doc.txt`.
- Abbott, I. H. & von Doenhoff, A. E., *Theory of Wing Sections*, Appendix IV.
- Van Ingen, J. (2008), The e^N method for transition prediction, AIAA 2008-3830.

A few commands come from XFOIL 6.99 and are not in the older primer: `ITER`, `INIT` in OPER, `CPWR`, `TFAC`, and `N` / `XTR` in VPAR. Type `?` in the relevant menu the first time to confirm them on your build.